

Doc 7: Linear Transformations

Module 1: Linear algebra | Part 1 of 3: Learn (theory)

1. A matrix is a machine that moves points

The rule $x \rightarrow A x$ takes a vector and returns a new vector. It is a **linear transformation**: $A(u + v) = A u + A v$ and $A(c v) = c A v$. In pictures: the **origin stays put**, straight lines stay straight, and an evenly spaced grid stays evenly spaced (but may be stretched, turned, or tilted). Example: $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ sends $(1, 1)$ to $(2, 3)$.

2. The columns tell the whole story

The standard basis vectors are $e_1 = (1, 0)$ and $e_2 = (0, 1)$. **$A e_1$ is column 1 of A and $A e_2$ is column 2.**

Every vector is $x = x_1 e_1 + x_2 e_2$, so $A x = x_1(\text{column 1}) + x_2(\text{column 2})$: the column view from Doc 5.

To build the matrix of a transformation: find where e_1 and e_2 land, and write those two points as the columns. Example: rotate 90° counter-clockwise. e_1 lands at $(0, 1)$ and e_2 lands at $(-1, 0)$, so the matrix is $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

3. A gallery of 2D transformations

Transformation	Matrix	e_1 lands at	e_2 lands at	det
Stretch by a and b	$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$	$(a, 0)$	$(0, b)$	ab
Reflect in the x-axis	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$(1, 0)$	$(0, -1)$	-1
Swap x and y	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$(0, 1)$	$(1, 0)$	-1
Rotate by angle θ	$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$	$(\cos \theta, \sin \theta)$	$(-\sin \theta, \cos \theta)$	1
Rotate 90° counter-clockwise	$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$	$(0, 1)$	$(-1, 0)$	1
Shear by k	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	$(1, 0)$	$(k, 1)$	1
Project onto the x-axis	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$(1, 0)$	$(0, 0)$	0

4. Doing several transformations in a row

Do T_1 first, then T_2 : the matrix is $A_2 A_1$. The first step sits closest to the vector, on the right. Order matters: rotate-then-stretch is not stretch-then-rotate.

Undoing a transformation: A^{-1} (Doc 6) reverses A . It exists only when $\det A \neq 0$, because a squashed shape cannot be un-squashed. **Determinant:** the factor by which A scales areas; negative means mirrored.

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Part 1 of 3: Learn (theory), continued

5. Basis and coordinates: one arrow, many descriptions

A **basis** is a set of vectors (two in 2D) that do not line up, so every vector can be built from them in exactly one way. The numbers used to build it are its **coordinates**. Coordinates depend on the basis; the arrow itself does not.

Example. $v = (3, 1)$ is $3 e_1 + 1 e_2$ in the standard basis. In the basis $b_1 = (1, 1)$, $b_2 = (-1, 1)$ it is $2 b_1 - 1 b_2$, because $2(1, 1) - 1(-1, 1) = (3, 1)$. Its coordinates in the new basis are **(2, -1)**.

Put the basis vectors in the columns of $B = [b_1 \ b_2]$. Then:

Direction	Formula	In words
Coordinates in basis $B \rightarrow$ standard	$v = B c$	multiply by B
Standard $\rightarrow$ coordinates in basis B	$c = B^{-1} v$	multiply by B^{-1} (Doc 6)

Change of basis changes the description, not the vector. Doc 8 will show that a good choice of basis (the eigenvectors) can make a transformation as simple as a plain stretch.

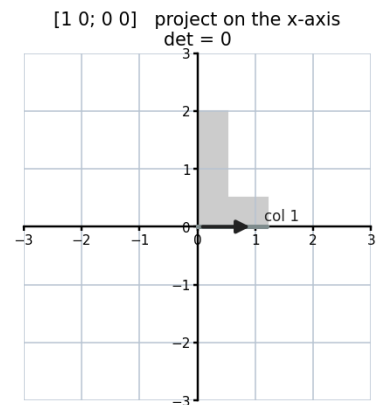
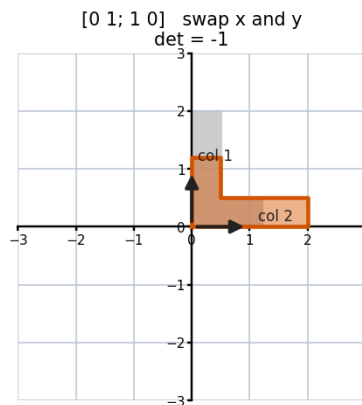
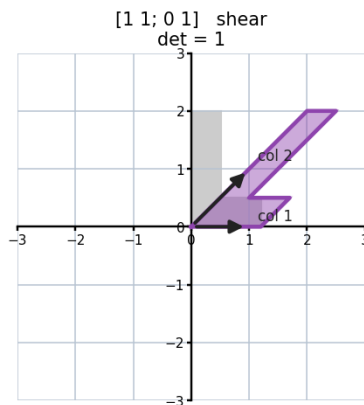
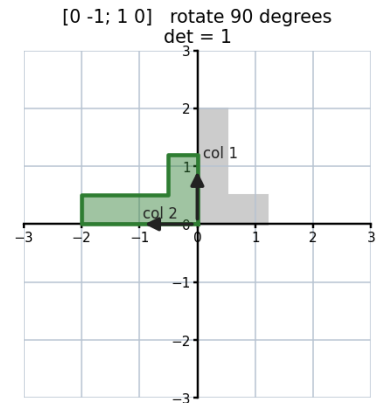
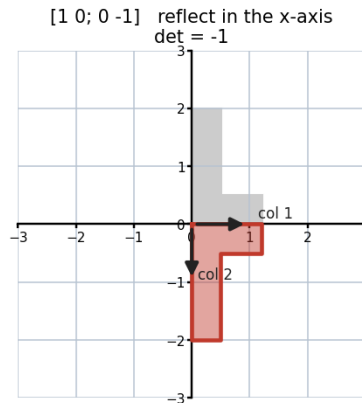
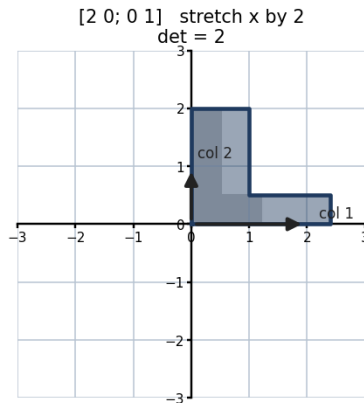
6. Where machine learning uses these ideas

- **Every layer** starts with a linear map $x \rightarrow W x$. A 3×2 matrix maps 2D inputs into 3D; a 2×3 matrix squashes 3D into 2D.
- **Attention** (Doc 29) makes queries, keys, and values by three different linear maps of the same input.
- **PCA** (Doc 9) finds a better basis for the data, one where the important directions come first.
- **Rotations:** rotary position embeddings rotate pairs of features by an angle that depends on the token's position.
- **Stacked linear layers** collapse into one matrix (Doc 5), so a network needs a nonlinearity between layers to gain power.

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Part 2 of 3: Worked examples. Every cell filled in. Read, then cover the answers and redo them.

Example 1: The gallery in pictures (grey = the original L, colour = the image)



Arrows show where e_1 and e_2 land: they are the columns of each matrix. The reflection, swap and projection change or destroy orientation; the rotation and shear keep areas.

Example 2: Building a matrix from where e_1 and e_2 land

		2
		1
0	-1	-1
1	0	2

Rotate 90° counter-clockwise: $e_1 \rightarrow (0, 1)$, $e_2 \rightarrow (-1, 0)$. Columns give $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.
Apply to $v = (2, 1)$: $2 \times (0, 1) + 1 \times (-1, 0) = (-1, 2)$. The grid gives the same answer.

Example 3: Transform many points at once

		1	0	1	2
		0	1	1	1
2	0	2	0	2	4
0	1	0	1	1	1

$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ stretches x by 2. Each column of the top matrix is a point: $(1, 0)$, $(0, 1)$, $(1, 1)$, $(2, 1)$.

One product transforms all four. This is how a layer processes a whole batch of inputs.

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Part 2 of 3: Worked examples, continued

Example 4: Order matters (rotate 90° and stretch x by 2)

Rotate first, then stretch: $S R$

		0	-1
		1	0
2	0	0	-2
0	1	1	0

Stretch first, then rotate: $R S$

		2	0
		0	1
0	-1	0	-1
1	0	2	0

$(S R) v$

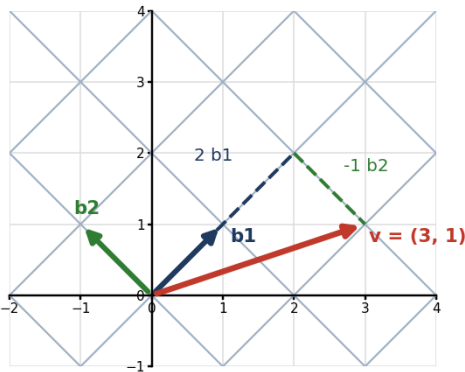
		1
		1
0	-2	-2
1	0	1

$(R S) v$

		1
		1
0	-1	-1
2	0	2

$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$. The matrix for R -then- S is $S R$ (the first step sits on the right). $S R = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}$ and $R S = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}$ differ, and so do the results on $v = (1, 1)$: $(-2, 1)$ versus $(-1, 2)$. Check: rotate $(1, 1)$ to $(-1, 1)$, then stretch x : $(-2, 1)$.

Example 5: Change of basis: $v = (3, 1)$ in the basis $b_1 = (1, 1)$, $b_2 = (-1, 1)$



$c = B^{-1} v$

		3
		1
0.5	0.5	2
-0.5	0.5	-1

Check: $B c = v$

		2
		-1
1	-1	3
1	1	1

$B = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ (columns b_1, b_2), $\det = 2$, $B^{-1} = 1/2 \times \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$. Coordinates $c = (2, -1)$: go 2 steps along b_1 , then 1 step back along b_2 , and you arrive at v .

Quick reference card

Columns of A	$A e_1 = \text{column 1}$, $A e_2 = \text{column 2}$	build a matrix from where e_1, e_2 land
Then	T_2 after T_1 has matrix $A_2 A_1$	first step on the right
Undo	A^{-1} (needs $\det A \neq 0$)	reverses the transformation
Area	$\det A = \text{area scale factor}$	negative = mirrored, 0 = squashed
Coordinates in basis B	$c = B^{-1} v$ and $v = B c$	columns of B are the basis vectors

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Part 3 of 3: Practice. Use a pencil; yellow cells are yours. Name: _____ Date: _____

A. Where do e_1 and e_2 land? Write the matrix

Transformation	A e_1 (column 1)	A e_2 (column 2)	Matrix A
Stretch x by 3 and y by 2			
Reflect across the y-axis			
Rotate 90° clockwise			
Shear: e_1 stays, e_2 moves to (2, 1)			
Project onto the y-axis			

B. Transform four points at once (columns of P are the points)

Shear $[1\ 2; 0\ 1]$ times P

		1	0	1	2
		0	1	1	-1
1	2				
0	1				

Rotation 90° $[0\ -1; 1\ 0]$ times P

		1	0	1	2
		0	1	1	-1
0	-1				
1	0				

Which of these two transformations changes areas? _____ (hint: find each determinant)

C. Order matters: swap x and y, and stretch x by 3

Swap first, then stretch: S W

		0	1
		1	0
3	0		
0	1		

Stretch first, then swap: W S

		3	0
		0	1
0	1		
1	0		

$S = [3\ 0; 0\ 1]$, $W = [0\ 1; 1\ 0]$.

Apply each to $v = (1, 2)$:

$S W v = (___, ___)$

$W S v = (___, ___)$

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Practice, continued

D. Determinants and area

Matrix	det	Effect on area
$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$		
$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$		
$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$		
$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$		
$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$		

E. Undo each transformation: find the inverse matrix

$\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$ Stretch	inverse	$\begin{bmatrix} & \\ & \end{bmatrix}$ inverse matrix
$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ Rotate 90° CCW	inverse	$\begin{bmatrix} & \\ & \end{bmatrix}$ inverse matrix
$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ Shear	inverse	$\begin{bmatrix} & \\ & \end{bmatrix}$ inverse matrix

Check the rotation: multiply $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ by your inverse. You should get the identity.

0	-1		
1	0		

Rotating by 90° and then by -90° brings every point home.

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Practice, continued

F. Change of basis

F1. Basis $b_1 = (1, 0)$, $b_2 = (1, 1)$, so $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$. Find the coordinates of $v = (5, 2)$, then check.

$c = B^{-1}v$

		5
		2
1	-1	
0	1	

Check: Bc should equal v

1	1	
0	1	

$c = (\quad , \quad)$

F2. Basis $b_1 = (2, 1)$, $b_2 = (0, 1)$. A vector has coordinates $c = (3, 4)$ in this basis. Find v in standard coordinates.

		3
		4
2	0	
1	1	

$v = (\quad , \quad)$

Standard $\rightarrow$ basis coordinates: multiply by _____.

Basis coordinates $\rightarrow$ standard: multiply by _____.

G. Challenge: two layers as two transformations

$W_1 = \begin{bmatrix} 1 & -1 \\ 2 & 0 \\ 0 & 1 \end{bmatrix}$ sends 2D points into 3D. $W_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ sends 3D points back to 2D. $x = (2, 3)$.

$W_1 x$

		2
		3
1	-1	
2	0	
0	1	

$W_2 W_1$

			1	-1
			2	0
			0	1
1	0	1		
0	1	1		

$(W_2 W_1) x$

		2
		3

Apply W_2 to your $W_1 x$ by hand: $(\quad , \quad)$. Does it match $(W_2 W_1) x$? _____

The outputs of W_1 all lie on a flat _____ inside 3D, even though W_1 has 3 rows.

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Answer key (page 1 of 2): check your work only after finishing

A. Matrices from basis images

Transformation	A e1	A e2	Matrix A
Stretch x by 3 and y by 2	(3, 0)	(0, 2)	[3 0; 0 2]
Reflect across the y-axis	(-1, 0)	(0, 1)	[-1 0; 0 1]
Rotate 90° clockwise	(0, -1)	(1, 0)	[0 1; -1 0]
Shear: e1 stays, e2 moves to (2, 1)	(1, 0)	(2, 1)	[1 2; 0 1]
Project onto the y-axis	(0, 0)	(0, 1)	[0 0; 0 1]

B. Four points at once

Shear times P

		1	0	1	2
		0	1	1	-1
1	2	1	2	3	0
0	1	0	1	1	-1

Rotation times P

		1	0	1	2
		0	1	1	-1
0	-1	0	-1	-1	1
1	0	1	0	1	2

det shear = 1, det rotation = 1. Neither changes areas.

C. Order matters

S W

		0	1
		1	0
3	0	0	3
0	1	1	0

W S

		3	0
		0	1
0	1	0	1
1	0	3	0

S W v = (6, 1): swap (1, 2) to (2, 1), then stretch x to (6, 1).
W S v = (2, 3): stretch to (3, 2), then swap to (2, 3). Different, so order matters.

D. Determinants and area

Matrix	det	Effect on area
[3 0; 0 2]	6	6 times bigger
[1 2; 0 1]	1	unchanged
[0 -1; 1 0]	1	unchanged
[1 0; 0 -1]	-1	unchanged, but mirrored
[2 4; 1 2]	0	collapses to a line

E. Inverses

0.5	0
0	0.25

stretch
inverse

0	1
-1	0

rotation
inverse

1	-2
0	1

shear inverse

Rotation times its inverse gives the identity.

		0	1
		-1	0
0	-1	1	0
1	0	0	1

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Answer key (page 2 of 2)

F. Change of basis

F1: c

		5
		2
1	-1	3
0	1	2

F1 check: $B c$

		3
		2
1	1	5
0	1	2

F2: $v = B c$

		3
		4
2	0	6
1	1	7

F1: $c = (3, 2)$, and $3(1, 0) + 2(1, 1) = (5, 2)$. F2: $v = (6, 7)$. Standard $\rightarrow$ coordinates: multiply by B^{-1} . Coordinates $\rightarrow$ standard: multiply by B .

G. Two layers as two transformations

W1 x

		2
		3
1	-1	-1
2	0	4
0	1	3

W2 W1

		1	-1
		2	0
		0	1
1	0	1	1
0	1	1	2

(W2 W1) x

		2
		3
1	0	2
2	1	7

W2 applied to $(-1, 4, 3)$ gives $(2, 7)$, which matches (W2 W1) x . $W2 W1 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ is a shear. The outputs of W1 lie on a flat plane inside 3D.

Key takeaways: a matrix moves points, and its columns are where e_1 and e_2 land; do T1 then T2 with A2 A1; det measures area scaling; A^{-1} undoes A; coordinates in a basis B are $c = B^{-1} v$, and $v = B c$ converts back.

Next: Doc 8 - Eigenvalues and eigenvectors.